Q.1 Why is ERP important in operations?

A. Improves UI design
B. Creates marketing campaigns
C. Manages social media
D. Integrates data and processes across departments

Q.2 Which of the following companies uses robotic assembly lines?

A. Tesla
B. Nestlé
C. McDonald's
D. Facebook

Q.3 How does price skimming work?

A. Prices are never changed
B. Prices increase over time
C. High price is set first, then gradually lowered
D. Low prices are set initially

Q.4 Which pricing model adjusts prices in real-time based on demand?

A. Psychological pricing
B. Bundle pricing
C. Dynamic pricing
D. Skimming

Q.5 A fitness app targets senior citizens with low-impact workouts. This is an example of:

A. Undifferentiated marketing
B. Behavioral segmentation
C. Niche positioning
D. Geographic segmentation

Q.6 What is a key benefit of Just-in-Time (JIT) inventory?

A. Long-term warranties
B. Reduced inventory holding costs
C. Higher storage costs
D. Early product launch

Q.7 Which type of pricing is often used by tech companies launching innovative products?

A. Price skimming
B. Cost-based pricing
C. Penetration pricing
D. Bundle pricing

Q.8 What does the "positioning" process aim to do?

A. Launch a new product
B. Determine pricing
C. Create a unique image in consumers' minds
D. Build distribution channels

Q.9 What is the role of feedback in the production process?

A. Increase output
B. Replace machines
C. Ensure more labor
D. Improve efficiency and quality

Q.10 The marketing strategy that addresses societal issues along with profit is called:

A. Societal marketing
B. Environmental marketing
C. Ethical marketing
D. Public relations

Q.11 A set of functions \left\{ {{f_1}(x),{f_2}(x),{f_3}(x),.....,{f_n}(x)} \right\} is said to be _____ on an interval I if there exist constants {c_1},{c_2},{c_3},.....,{c_n} not all zero, such that {c_1}{f_1}(x) + {c_2}{f_2}(x),.....,{c_n}{f_n}(x) = 0,\;\;\forall x \in I

A. linear independent
B. linear dependent

Q.12 {\text{The}}\,\,{\text{input}}\,\,{\text{function}}\,\,g(x)\,\,{\text{can}}\,\,{\text{have}}\,\,{\text{one}}\,\,{\text{of}} function}}\,\,k;\,\,{\text{A}}\,\,{\text{polynomial}}\,\,{\text{function;}}\,\,_____.

A. {{(I)}}\,\,\,\,\,\,\,\,\,\,\,\,{{An}}\,\,{{exponential}}\,\,{{function}}\,\,{e^{ x}}
B. {{(II)}}\,\,\,\,\,\,\,\,\,\,\,\,{{The}}\,\,{{trigonometric}}\,\,{{functions}}\,\, x,\,\, x
C. {{(IV)}}\,\,\,\,\,\,\,\,\,\,\,\,{{All}}\,\,{{(I),}}\,\,{{(II)}}\,\,{{and}}\,\,{{(III)}}{{.}}
D. {{(III)}}\,\,\,\,\,\,\,\,\,\,\,\,{{Finite}}\,\,{{sums}}\,\,{{and}}\,\,{{products}}\,\,{{of}}\,\,{{these}}\,\,{{functions}}

Q.13 \begin{gathered} {\text{Consider}}\,\,{\text{the}}\,\,{\text{non - homogeneous}}\,\,{\text{linear}}\,\,{\text{differential}}\,\,{\text{equation,}}\,\,{a_n}{y^{(n)}} + {a_{n - 1}}{y^{(n - 1)}} + ... + {a_1}y' + {a_0}y = g(x).\,\, \hfill \\ If\,\,{\text{the}}\,\,{\text{input}}\,\,{\text{function,}}\,\,g(x) = {e^{5x}},\,\,{\text{then}}\,\,{\text{the}}\,\,{\text{assumed}}\,\,{\text{particular}}\,\,{\text{solution}}\,\,{y_p}\,\,{\te \hfill \\\\ \end{gathered}

A. A{e^{5x}} 5x + B{e^{5x}} 5x
B. (Ax + B){e^{5x}}
C. (A{x^2} + Bx + C){e^{5x}}
D. A{e^{5x}}

Q.14 The differential operator L is called an annihilator operator of the function f if: _____.

A. (iv) All (i), (ii) and (iii).
B. (ii) y = f(x) defines a sufficiently differentiable function.
C. (i) L is a linear differential operator with constant coefficients.
D. (iii) The function f is such that L(y)=0.

Q.15 {\text{If}}\,X\,{\text{is}}\,\,{\text{the}}\,\,{\text{remaining}}\,\,{\text{amount}}\,\,{\text{of}}\,{\text{the}}\,\,{\tex = kX,\,{\text{where}}\,\,k < 0\,\,{\text{because}}\,X\,\,{\text{is}}\,\,_____.

A. {{decreasing}}
B. {{increasing}}

Q.16 \begin{gathered} {\text{Consider}}\,\,{\text{the}}\,\,{\text{non - homogeneous}}\,\,{\text{linear}}\,\,{\text{differential}}\,\,{\text{equation,}}\,\,{a_n}{y^{(n)}} + {a_{n - 1}}{y^{(n - 1)}} + ... + {a_1}y' + {a_0}y = g(x).\,\, \hfill \\ If\,\,{\text{the}}\,\,{\text{input}}\,\,{\text{function,}}\,\,g(x) = (9x - 2){e^{5x}},\,\,{\text{then}}\,\,{\text{the}}\,\,{\text{assumed}}\,\,{\text{particular}}\,\,{\text{solution}}\,\,{y_p}\,\,{\ \hfill \\\\ \end{gathered}

A. (Ax + B){e^{5x}}

B. A{e^{5x}}

C. (A{x^2} + Bx + C){e^{5x}}

D. A{e^{5x}} 5x + B{e^{5x}} 5x

Q.17 The homogeneous linear differential equations always possess the _____ solution.

A. trivial

B. non-trivial

Q.18 In differential equations, the symbol D is known as _____ operator.

A. relational

B. differential

C. arithmetic

D. integral

Q.19 The polynomial linear differential operators with constant coefficients _____.

A. identical

B. distributive

C. commute

D. inverse

Q.20 \begin{gathered} {\text{A}}\,\,{\text{function}}\,\,{\text{satisfying}}\,\,{\text{the}}\,\,{\text{differential}}\,\,{\text{equation}}\,\,{\t \hfill \\ {\text{that}}\,\,{\text{the}}\,\,{\text{slope}}\,\,{\text{of}}\,\,{\text{the}}\,\,{\text{curve}}\,\,{\text{at}}\,\,{\text{th \hfill \\\ \end{gathered}

A. {{boundary}}

B. {{initial}}

Q.21 The simplest examples of periodic functions from \mathbb{R} into \mathbb{R} included the well-known Sine and Cosine functions, since for each k \in \mathbb{Z}

A. False

B. True

Q.22 A Fourier series expansion of a function is a representation of the function as a linear combination of sines and cosines as _____

A. \{ 1, nx, nx\} _{n = 0}^

B. \{ 1, nx, nx\} _{n = 1}^

C. \{ 1, nx, nx\} _{n = - }^

D. \{ 1, nx, nx\} _{n = 0}^{ - }

Q.23 \begin{gathered} {\text{If a function}}\,f\left(x \right){\text{, having the derivatives of all orders on some neighbourhood }}N\left(0 \right){\text{ of }}0{\text{, }} \hfill \\ {\text{be chosen first and the Taylor series }}\sum\limits_{n = 0}^\infty {\frac{{{f^{\left(n \right)}}\left(0 \right)}}{{n!}}} {x^n}\,\,{\text{be constructed, }} \hfill \\ {\text{then this power series will have }}f\left(x \right){\text{ as its sum on N}}\left(0 \right). \hfill \\\ \end{gathered}

A. False

B. True

Q.24 Let f be defined by a power series with finite radius of converges R. If \sum\limits_{n = 0}^\infty {{{(- 1)}^n}{a_n}{R^n}} diverges, then \mathop {\lim }\limits_{x \to ({x_0} + R) + } f(x) = \sum\limits_{n = 0}^\infty {{{(- 1)}^n}{a_n}{R^n}}

A. True

B. False

Q.25 \mathop {\lim }\limits_{n \to \infty } \frac{{{r^n}}}{{n!}} =

A. = 0, 0 < r <

B. = n! , 0 < r <

Q.26 \begin{gathered} {\text{If a function is defined in neighbourhood of 0 as ; }}f\left(x \right) = \left\{ \begin{gathered} {e^{ - \frac{1}{{{x^2}}}}},\,\,x \ne 0 \hfill \\ 0,\,\,\,\,\,\,\,\,\,x = 0 \hfill \\\ \end{gathered} \right.{\text{,}} \hfill \\ {\text{then Taylor series of }}f{\text{ about 0 is - - - - - - - }}{\text{.}} \hfill \\\ \end{gathered}

A. _{n = 0}^ {{{{f^{(n)}}({0 + })}}{{(n)!}}{x^n}}

B. _{n = 0}^ 0

Q.27 \tan x is an odd function, its derivatives of even order are equal to

A. at x_0=0

B. x x at x_0=0

C. 1 at x_0=0

D. 0 at x_0=0

Q.28 A given function " f(x) " can be represented as the sum of a convergent power series in " x ", if f(x) must have (be) --------- in the neighborhood of x=0.

A. continuous

B. derivatives of finite order

C. derivatives of all order

D. extreme points

Q.29 If f (x) and g(x) are periodic with the _____ \omega, then h(x) = af (x) + bg(x) is also a periodic function with period \omega.

A. cosine

B. sine

C. same period

D. different period

Q.30 A function f:\Omega \subset \mathbb{R} \to \mathbb{R} is said to be periodic if there exists a nonzero real number w such that _____ .

A. f(x) = f(x - w),x

B. f(x) = f(x w),x

C. f(x) = f(x + w),x

D. f(x) = f(x w),x
